

Machine Learning AoL Proposal

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Project Title and Problem Statement

a. Project Title

Used Laptop Price Predictor: A Machine Learning Approach to Fair Market Valuation

b. Problem Statement

Determining the market value of a used laptop is difficult because of the decrease in value of hardware and the vast variety of specifications available. People find it difficult to price laptops based on the CPU, GPU, RAM, and storage type. We plan to build a classical machine learning regression model that predicts the market price of a laptop based on its specifications. By deploying this predictor, we hope to help people make the correct decision in buying or selling their secondhand laptop.

Dataset Source and Description

a. Dataset Source

The dataset we're planning to use for this paper is from Kaggle with the title of Laptop Price Prediction Dataset by Jackson Divakar, the link can be accessed at:

[Laptop Price Prediction Dataset | Kaggle](#)

b. Description

This dataset contains a comprehensive collection of laptop information designed for exploring and predicting laptop prices based on their specifications. It includes data on various laptop brands, processors, RAM, storage, graphics, display features, operating systems, and warranty details.

Selected ML Methods and Baseline

a. Selected ML Methods

As for our machine learning method, we will use Random Forest Regression because it's very robust so it can capture non-linear relationships between each laptop's specifications

and the final price. This algorithm basically builds hundreds of decision trees and each tree is built using random and different specifications of the laptop and the features. We will also use Support Vector Regression (SVR) because it works well when the relationships between specs and price is complex (not just a straight linear line). Since the laptop datasets contain many categorical and numerical features like brands, CPU, GPU, RAM, etc, SVR works well to map this complex high-dimensional data.

b. Baseline

Our baseline model will be a Multiple Linear Regression model. This will provide a fundamental performance metric to evaluate whether more complex ensemble models like Random Forest Regression and Support Vector Regression resulting in the prediction accuracy.

Planned Evaluation Metrics

a. Mean Absolute Error

MAE is used to measure the average of the error in the price predictions. It helps check how wrong/right the predicted prices are.

b. Root Mean Squared Error

RMSE is used to penalize large prediction errors more heavily, effectively preventing outliers. This is important because overestimating or underestimating a laptop's price, especially if it's a high end laptop, by a big difference is unwanted.

c. R-squared

R-squared is used to understand the proportion of variance in laptop prices that is predictable from the hardware features.

Deployment and User Testing Plan

a. Deployment

The trained Machine Learning model will be deployed as a web application. The app will prioritize a user-friendly interface so users can input the specifications of a laptop and receive almost instant price prediction.

b. User Testing Plan

We will control usability testing with a minimum of 5 users. Users will be given specific usage scenarios, such as evaluating their personal laptops or pricing a hypothetical

marketplace listing. After using the app, they will be told to give feedback like ease of use and accuracy, User Interface and output interpretability.

References

Siburian, A. D., Sitompul, D. R. H., Sinurat, S. H., Situmorang, A., Ruben, Ziegel, D. J., & Indra, E. (2022). Laptop Price Prediction with Machine Learning Using Regression Algorithm. JUSIKOM PRIMA (Jurnal Sistem Informasi dan Ilmu Komputer Prima), 6(1), 87-91.